

Instructions

CS 6650 – Final Project Report

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1 Instructions

This document explains how to build and run the Manager–Worker distributed monitoring system. The system supports any number of managers (minimum 3 for Raft) and any number of workers.

2 Requirements

- Linux environment (Khoury cluster)
- g++ with C++17 support
- make
- Bash shell

3 Build the Project

From the project directory, run:

```
make clean  
make
```

This produces two executables: `manager` and `worker_bin`.

4 Running Managers

4.1 General Syntax

```
./manager <node_id> <raft_port> <peer_list>
```

- `<node_id>` — unique ID (e.g., `node1`)
- `<raft_port>` — unique RAFT port (e.g., `5001`)
- `<peer_list>` — comma-separated list of other managers

4.2 Example: Single Manager

```
./manager node1 5001 ""  
(Yes, the quotes are important)
```

4.3 Example: Three Managers

```
./manager node1 5001 node2:5002,node3:5003
./manager node2 5002 node1:5001,node3:5003
./manager node3 5003 node1:5001,node2:5002
```

4.4 Example: Four Managers

```
./manager node1 5001 node2:5002,node3:5003,node4:5004
./manager node2 5002 node1:5001,node3:5003,node4:5004
./manager node3 5003 node1:5001,node2:5002,node4:5004
./manager node4 5004 node1:5001,node2:5002,node3:5003
```

4.5 Example: N Managers

Each manager must:

1. Have a unique node ID
2. Have a unique RAFT port
3. List all other managers in its peer list

5 Running Workers

5.1 General Syntax

```
./worker_bin <worker_id>
```

5.2 Example

```
./worker_bin worker1
./worker_bin worker2
./worker_bin worker3
```

6 Cluster State Output

Worker state is written by the leader to `workers.json`, updated every few seconds.

```
{
  "workers": [
    {
      "id": "worker1",
      "online": true,
      "last_seen_seconds_ago": 1,
      "status": "active"
    }
  ]
}
```

7 Automated Failover Testing

Run the test script:

```
chmod +x run_tests_auto.sh
./run_tests_auto.sh
```

It demonstrates leader election, failover, quorum loss, and recovery.

8 Stopping All Processes

```
pkill -f "./manager"
pkill -f "./worker_bin"
```

9 Logs

Manager logs:

```
node_<node_id>.log
```

Worker logs:

```
<worker_id>.log
```

View logs:

```
tail -n 50 node_node1.log
```

10 Recommended Configuration

- At least 3 managers for Raft
- Prefer odd numbers (3, 5, 7)
- Any number of workers may be used

Managers and workers can be added or removed dynamically, and the cluster maintains eventual consistency.